SAINT MICHAEL COLLEGE OF CARAGA

THE CRISIS MANAGEMENT PREPAREDNESS AND PRACTICES OF HOTELS IN BUTUAN CITY

A Thesis Presented To

The faculty of

College of Tourism and Hospitality Management

In Partial Requirements for the Degree

BACHELOR OF SCIENCE IN HOSPITALITY MANAGEMENT

By

KIRSTEEN BLESS G. HERMOSADA

JENNIFER P. DEMATA

CRISTINE V. LIGCUBAN

VINCENT REY C. MAGALLANO

EARL MAURICE A. MAQUIRATO

JORGE REVIN E. TAMAYO

ADVISER: DOMINIQUE D. ABARQUEZ

CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

The hospitality sector is highly susceptible to various crises such as natural disasters, public health emergencies, and operational disruptions; only a few researches have been conducted that detail how hotel crisis management is operated in small cities. According to Ritchie (2004), crisis management entails preparations before a crisis occurs, including risk assessments, formulating crisis management plans, training employees, and monitoring of threats. In this light, Faulkner (2001) has suggested that crisis response involves the activation of crisis teams, effective communication, and deployment of emergency procedures that can minimize disruptions. It is for this reason that in the growing hotel industry of Butuan City, it remains critical to determine how these approaches are implemented to ensure guest safety and organizational resilience. However, hardly any observed evidence is found regarding the level of crisis preparedness and response among hotels in small urban areas. Thus, this study tries to investigate the crisis management practices in hotels within Butuan City to add to the growing body of knowledge on the subject matter and support future improvements in local hotel crisis preparedness.

Cross-national studies examining how the hospitality sector responds to crises such as pandemics, natural disasters, and economic downturns clarify the concepts of pre-crisis management, resilience, and group communication. Tiwari (2024) asserts that hotels that employ proactive planning and skillfully implement crisis response techniques will be more competitive and resilient due to their ability to mitigate economic disruptions. Similarly, Belias and Trihas (2022) emphasized that effective human resource

training during crises, particularly for front-office employees, enhances adaptability, maintains service quality, and strengthens a hotel's competitive advantage by ensuring that staff can

cope with operational changes and safety challenges. Hotels that are creative, adaptable, and good at connecting with guests are more likely to be successful and get repeat business. According to Wut et al. (2021), hotels can remain strong by generating new ideas that enable them to adapt to emerging business models, technologies, and service delivery methods. Saputra et al. (2022) also found that hotels could stay open during uncertain times because they had flexible staff, contactless services, and technology. Wu et al. (2021) observed that employee engagement and feedback improve preparedness for crises. On the other hand, Muriuki et al. (2024) and Casal-Ribeiro et al. (2023) stated that effective communication is crucial for calming people down and building trust. Gáll, Chlpek, and Frasch (2024) stated that the COVID-19 pandemic caused significant problems for the international economy. To address this, Adi Saputra et al. (2022) proposed data-driven, adaptive, and sustainable crisis management measures to enhance future preparedness for the hotel sector. Expanding this view, Alsawidan et al. (2024) advocate for integrated frameworks linking governments, private sectors, and communities—affirming that adaptive leadership and cross-sectoral cooperation are central to effective crisis resilience and recovery. Research conducted locally in the Philippines has demonstrated the importance of readiness and adaptability in maintaining stability in the travel and hospitality industry during emergencies. According to Ochoa (2024), most businesses believe they can recover from calamities that could impact their operations and bottom line. Similarly, Carreon et al. (2022) emphasized that disaster preparedness should take center stage among hotel issues and should be incorporated into all hotel operational plans. Liwanag et al. (2024) noted that continuous crisis management training is crucial for hotel receptionists to effectively manage hazards and reduce risks. Resilience is also predicated on strong leadership, carefully thought-out crisis

planning, efficient communication, high-quality marketing, and government coordination, according to Cabaguing et al. (2024). Adem and Ylagan (2022) noted that hotels in the National Capital Region (NCR) weathered the pandemic by complying with government safety protocols and utilizing government assistance programs. On the same note, Lacerna and Dalugdog (2025) explained that San Juan Beach resorts were better equipped to handle emergencies but also needed to improve their supply management and evacuation programs further. Apple Sangat and Iris Estores (2023) agreed that some resorts are prepared for the risks but continue to have gaps in communication with staff and stakeholders. Likewise, Lopez Jr. et al. (2022), Gaudiel (2023), and Basilio (2025) found that socioeconomic factors, coordination, and awareness play significant roles in affecting both disaster preparedness and community resilience.

The study will determine the capacity of city hotels to respond to crises and prepare for them in terms of ensuring safety and readiness for hotel employees and guests. By pinpointing areas to be improved and gaps in current practices, this research will enable hotel management to revise more effective crisis plans, enhance employee training programs, and ensure guest safety during emergencies. The results shall serve as reference for the local hospitality industry and may be used by legislators, hotel managers, and owners to draft and amend policies to further support safer and more resilient tourism operations in Butuan City. Ultimately, this study aims to enhance the protection and welfare of hotel employees and guests, as well as contribute to sustainable growth and operational resiliency. It provides an opportunity for practical improvements in hotel policies, staff training, and crisis protocols and contributes to the academic literature on crisis management in the hospitality industry, especially in the Philippines. By providing a suggested crisis management model or training module, hotels will have a realistic means to improve preparedness and guest security whenever a crisis event occurs in the future.

The study will examine the preparation and response of hotels in Butuan City for emergency scenarios, such as natural disasters, public health crises, or operational disruptions that threaten the safety of hotel personnel and guests while maintaining a smooth business operation. It identifies the importance of staff training, formal crisis management programs, and collaboration with public agencies. The study also expands on previous knowledge that in enhancing resilience, leadership, government support, and employee competencies are crucial in the hospitality industry. These results will contribute to the design of better practices for hotel managers, policymakers, and the tourism industry to advance preparedness and emergency response strategies with long-term benefits to protect and ensure a safe environment for employees and guests alike.

1.2 Theoretical Framework

The research builds on three popular theories: Crisis Management Theory, Resilience Theory, and Situational Crisis Communication Theory (SCCT). These theories describe the process of crisis preparation, management, and recovery at hotels. When combined, they help us realize that hotels will be prepared, flexible, and communicative in the event of an issue.

Crisis Management Theory

According to (Fink, 1986; Mitroff, 1994), crises typically progress through the following four stages: warning, acute (also referred to as intense), chronic (sometimes referred to as long-term), and resolution. In line with this theory, firms can reduce the impact of a crisis on their operations by ensuring they have adequate planning, risk management, and personnel preparedness. In the hospitality sector, this means ensuring that all employees, especially those at the front desk—the first point of contact with visitors—are prepared to act appropriately and respond immediately in an emergency. Hotels are prone to numerous risks, including health risks, natural disasters, accidents, and reputational damage. The Theory of Crisis Management emphasizes the importance of preparation for such risks. By utilizing backup plans, conducting simulated drills, and ensuring effective communication, hotels can better protect their customers and employees and respond more effectively in emergencies. Backing up, simulating, and ensuring the use of information and communication channels enable hotels to maintain the security of guests and employees and to recover more quickly in the event of an emergency.

Resilience Theory

Resilience Theory is about how well a group can handle and recover from setbacks. This theory not only examines what happens during a crisis but also considers how groups recover and become stronger in the aftermath. In a hotel, being resilient can mean responding to a health problem by making the rooms cleaner, allowing people to eat in their rooms, or adjusting the way guests move through common areas. It may entail minimizing expenses, optimizing resource utilization, and maintaining critical services despite a decline in visitors during a weak economy. Being resilient involves more than just surviving; it also entails growing, adapting, and preparing for the future, According to Holling, 1973.

Situational Crisis Communication Theory (SCCT)

SCCT, developed by Timothy L. Coombs in 2007, focuses on how organizations should communicate during crises. This theory suggests that an organization's communication strategy depends on the type of crisis and its perceived level of responsibility. In the hospitality industry, people at the front desk become critical — they are the hotel's direct link to guests when something goes wrong. If staff can remain calm, provide trustworthy information, and demonstrate honesty, they help guests feel confident in the hotel. Effective communication during a service interruption or illness can reduce guest anxiety and protect the hotel's reputation. In summary, these three theories work together to provide a comprehensive understanding of how hotels should handle crises. Crisis Management Theory emphasizes the steps before and during the crisis (planning and readiness). Resilience Theory emphasizes recovery and adaptation after the crisis. SCCT emphasizes how communication affects both the trust of guests and the hotel's image. When hotels strike a balance between preparation, flexibility, and clear communication, they protect their guests and staff, and maintain operations—even in uncertain times.

1.3 Review of Related Literature

This section has reviewed those literature and research works that substantiate the core theme of the study. A good company image ensures success, particularly in the hospitality industry, where the image of hotel staff and guests has a mutual influence on the assessment of the quality and safety of service. The role of frontline staff like receptionists, housekeeping staff, and food service staff is very important in the formation of the first impression among customers about the standards and preparedness of the hotel. While most employees perceive that they are prepared to handle any emergency, the actual situations bring out the weaknesses in training, coordination, and response mechanisms during crises. At the same time, the experiences of guests reflect whether these preparedness measures were well implemented. Identification of potential threats and responding confidently remains crucial for any successful crisis management. Recent studies indicate that hotel management has given more importance nowadays not only to guest safety but also to ensuring the well-being of the employees so that the staff and customers feel safe within the hospitality environment.

Sex

Sex is usually identified as male or female, and this is taken into consideration in research because differences may influence the subjects under study, either in behavior or response, due to biological and social differences. Across these studies, gendered leadership tendencies emerged: female leaders emphasized empathy, communication, and staff welfare, while male leaders prioritized cost control, strategy, and performance (Goll et al., 2025; Zhang et al., 2023; Mensah, 2021).

Goll et al. (2025) examined micro-level leadership during the pandemic and found that adaptability, emotional intelligence, and transparent communication enhanced team cohesion and resilience. Their

findings indicate that leaders who combined empathy with decisive action were better able to manage operational uncertainty. Similarly, Zhang et al. (2023) analyzed hotels in China and Macau, emphasizing the influence of institutional support on crisis outcomes. Hotels embedded in well-structured governance systems demonstrated stronger recovery and performance due to clearer protocols and coordinated responses. Mensah's (2021) research in Ghana focused on the psychological experiences of hotel employees, showing that empathetic leadership improved motivation, trust, and commitment despite heightened stress. Collectively, these findings highlight that effective crisis leadership in hospitality requires both strategic foresight and emotional awareness. The ability to connect with employees while implementing adaptive strategies proved vital to organizational resilience. As demonstrated across multiple contexts, leadership that integrates empathy, adaptability, and institutional alignment offers the most sustainable response to crises such as COVID-19.

Age

Age significantly influences individuals' crisis preparedness and response, particularly in the hospitality industry where composure and adaptability are essential. Chen and Cong (2023) found that older adults demonstrate higher levels of preparedness due to accumulated experience, a stronger sense of responsibility, and more effective coping mechanisms. Their study explained that older workers' prior exposure to challenges enables them to stay composed under stress, make sound judgments, and manage uncertainty more effectively. This experience-based confidence allows them to take safety actions and maintain operational stability during crises. In contrast, younger hotel employees often show greater fear and psychological strain when facing unpredictable events, reflecting limited exposure to crisis situations and underdeveloped coping skills (Chen & Cong, 2023; Jung et al., 2024). Jung et al. (2024) observed that

although younger workers are technologically skilled and adaptive, they experience higher anxiety levels, rely more heavily on guidance, and struggle to regulate emotions under pressure.

Meanwhile, older employees remain calmer and more resilient, often taking initiative to support and guide younger colleagues, thereby enhancing team coordination and morale (Chen & Cong, 2023; Jung et al., 2024). These differences suggest that age shapes both psychological readiness and behavioral responses during crises, influencing the overall resilience of hospitality organizations. Older employees' emotional stability and experience complement younger workers' flexibility and innovation, forming a balanced workforce capable of responding effectively to uncertainty. Encouraging intergenerational collaboration allows organizations to support each group's strengths, wisdom and preparedness from older employees and agility and creativity from younger ones to improve preparedness and recovery strategies. Thus, age should be regarded not merely as a demographic variable but as an element of crisis competence and resilience, influencing how individuals perceive risk, respond under pressure, and contribute to organizational stability (Chen & Cong, 2023; Jung et al., 2024).

Length of Service

Digital transformation plays a crucial role in enhancing service quality, operational efficiency, and employee adaptability. Existing research highlights that digitalization not only enhances organizational performance but also strengthens employees' crisis preparedness, satisfaction, and customer orientation (Leo et al., 2022; Medina-Garrido et al., 2023). Leo et al. (2022) emphasized that technological integration supports employees' ability to manage workload demands and maintain service consistency under pressure, encouraging confidence and proactive problem-solving. Likewise, Medina-Garrido, Biedma-Ferrer, and Bogren (2023) found that digital transformation contributes to employees' psychological

resilience and satisfaction, particularly when coupled with strong organizational and institutional support. Longer-tenured workers tend to exhibit greater adaptability, utilizing experience and technological familiarity to respond effectively to operational disruptions (Leo et al., 2022; Medina-Garrido et al., 2023).

Furthermore, work family policies and organizational backing play a moderating role by improving employees' well-being and commitment, especially among long-serving staff, thereby reducing turnover intentions and stress during crises. Both studies collectively suggest that technological adaptation when aligned with supportive organizational practices enhances individual and collective preparedness, reinforcing workforce stability and performance in crisis-prone industries such as hospitality and tourism. Hence, digitalization should be viewed not only as an operational necessity but also as a strategic enabler of employee resilience, engagement, and preparedness, promoting long-term organizational sustainability and effective service delivery during uncertain times (Leo et al., 2022; Medina-Garrido et al., 2023).

Position

It is much easier for employees to react promptly, collaborate effectively, and maintain order in the event of unanticipated events when roles and responsibilities are well defined. In addition to improving readiness in unpredictable times, strategic staff placement encourages cooperation and communication amongst all parties. As (Saflow et al., 2023) note, employees should know exactly what they are supposed to do in case of an emergency. Each team member plays a role in the team's planning and quick action. They put much effort into developing such teams, enabling them to create specialized teams capable of handling a broad range of events and situations. As (Jaaffar et al. 2023a) have noted, the hierarchy and leadership of hotel employees are essential in a crisis to ensure stability. However, both

papers indicate that it is paramount for each team member to have a clear, accurate understanding of their specific responsibilities. This will help them to relate well, be prepared, and respond to the potential situation. This organization comprises both leaders and field personnel.

Educational Attainment

A quantitative descriptive study in Tagum City, Philippines, examined knowledge, practice, and emergency and disaster preparedness among barangay responders. Cardinal and Lagura (2023.) found that respondents' knowledge and preparedness for managing emergency and disaster situations were moderate. Their research found that barangay responders who were more knowledgeable and aware of disaster procedures were more likely to exhibit effective preparedness behaviors. The findings also indicated that ongoing education and training greatly enhance readiness and response effectiveness. This highlights the importance of local initiatives in disaster risk mitigation and building local capacity (Cardinal & Lagura, 2023). These findings underscore the importance of localized knowledge and preparedness among frontline responders in enhancing community resilience.

Kalın and Yeşilyurt (2025) also found a direct correlation between students' perceptions of disaster risk and their readiness. The research indicated that students who were aware of the dangers related to catastrophes were sufficiently prepared and actively responded during emergencies. This suggests that people's awareness of risks is directly related to their readiness for disasters (Kalın & Yeşilyurt, 2025). The two essays emphasize the interdependence of awareness, perception, and readiness, suggesting that disaster management is based on ongoing education and practice. Enhancing knowledge and risk perception among various groups—from barangay responders to student societies—can help them become better prepared, less at risk, and more resilient to natural and artificial hazards. In general,

these results indicate that education, awareness, and effective risk communication are key to achieving sustainable disaster preparedness and response.

Crisis readiness

Crisis readiness is a multidimensional organizational capability shaped by leadership, strategic planning, human resources, communication systems, learning mechanisms, and adaptive cultures that enable institutions to anticipate, respond to, and recover from crises effectively. Mugyenyi and Heisi (2025) demonstrate that emotional and psychological readiness during health emergencies is deeply shaped by organizational climate, leadership support, communication systems, and institutional culture. Their review highlights that preparedness improves when organizations institutionalize trust, collaboration, clear protocols, and psychosocial support. Similarly, Malhooz (2025) frames crisis readiness as a holistic organizational process, integrating strategic foresight, operational preparedness, and human factors. She argues that readiness is strengthened when leadership, transparent communication, training, and learning systems operate together before, during, and after crises. Abeykoon and Perera (2025) show that in the hospitality sector, HRM readiness—particularly training, workforce adaptability, and technology acceptance significantly influences organizational responsiveness during disruptions. This aligns with Burton et al. (2025), who stress that outbreak readiness depends on multi-sector partnerships, workforce competence, and proactive planning, emphasizing preparedness as a shared responsibility across institutions. Galaura (2024) extends this perspective to the public sector, finding that organizational development, leadership agility, and strategic planning enhance readiness in post-pandemic environments. Likewise, Anwari and Suzianti (2025) reveal that integrating knowledge management with enterprise risk and crisis management improves organizational preparedness by strengthening learning, leadership effectiveness, and operational resilience. In supply chain contexts, Ramos et al. (2024) argue

that crisis readiness is driven by integration, flexibility, and adaptive capacity, enabling organizations to absorb shocks and sustain performance during disruptions. Finally, Björck et al. (2025) highlight that design thinking approaches, such as scenario mapping, empathy, and rapid prototyping, enhance crisis communication readiness by improving sense-making and coordinated responses in complex crises.

Policy and Plans

The studies collectively emphasize that effective policy outcomes depend on the alignment of planning instruments, governance structures, and institutional capacity over time, framing planning as a dynamic, coordinated process rather than a static policy document.

At the national development level, Song et al. (2025) show that China's low-carbon transition has been effectively driven by targeted industrial policies embedded in Five-Year Plans. Their findings highlight that planning continuity, sector-specific targeting, and strong governance integration allow national plans to act as practical instruments of structural change rather than symbolic policy statements. This emphasis on structural alignment resonates strongly with climate adaptation research. Keenan, Ajibade, and Tietjen (2025) show that although human mobility is increasingly acknowledged in National Adaptation Plans, it remains weakly operationalized because institutional capacity, land governance, and justice considerations are often overlooked. Their findings suggest that adaptation policies fail when planning frameworks do not engage with underlying governance constraints. Similarly, Rege and Agbolahon (2024) argue that forest and landscape restoration policies underperform when treated as isolated projects rather than integrated planning processes. Both studies emphasize that environmental policy effectiveness requires cross-sectoral coordination and attention to social and institutional dimensions, not just technical solutions. At the urban and sectoral planning level, Longato et al. (2024) reinforce this

argument by demonstrating that failures in implementing nature-based solutions stem from poor selection and coordination of policy instruments rather than lack of ambition. Their policy-instrument framework highlights the need to align regulatory, economic, and informational tools with planning objectives. This concern with instruments–context mismatch parallels Alizadeh (2025), who identifies a persistent gap between development plan objectives and institutional realities in Iran's higher education system. Alizadeh shows that centralized governance and weak enforcement mechanisms undermine financial autonomy, underscoring that planning outcomes depend on enforceable governance arrangements, not policy intent alone. The importance of integrating long-term vision with operational mechanisms is further developed by Malik, Szewczuk-Stępień, and Bebenek (2025). Their study reframes sustainable development policy as an integrated resilience planning process, arguing that long-term strategies must be directly connected to short-term action plans. This approach aligns closely with findings from crisis management literature, which consistently emphasize the dangers of fragmented and reactive planning. Within the hospitality and tourism sector, Cabaguing et al. (2024), Tiwari (2024), and Hazaa et al. (2021) collectively argue that effective crisis management in hospitality and tourism depends on institutionalized, continuously updated planning rather than ad hoc actions. Their studies emphasize embedding crisis protocols into formal plans, strengthening governance, leadership, and communication, and treating crisis management as an ongoing planning cycle within organizational and policy structures. Additionally, This planning cycle perspective is further elaborated by Casal-Ribeiro et al. (2023), whose meta-analysis shows a shift in tourism and hospitality toward proactive, multi-stage crisis management frameworks encompassing prevention, response, recovery, and learning. Their work positions crisis planning as a long-term resilience policy tool, aligning closely with Malik et al.'s emphasis on linking strategy with action across time horizons. Leadership and institutional capacity emerge as cross-cutting

enablers across sectors. Jaaffar et al. (2023) demonstrate that transformational leadership significantly enhances the implementation of crisis management plans in hotels, improving coordination and performance during crises. This finding aligns with Lytje and Dyregrov (2025), who show that bereavement policies in education only improve preparedness when supported by leadership commitment, training, and local ownership. Both studies highlight that policies fail when they are detached from the actors responsible for implementation, reinforcing the human dimension of planning. Finally, Magno et al. (2025) show that epidemiology serves as a strategic policy tool in Brazil's health planning, highlighting that evidence-based approaches, institutional capacity, and research–policy alignment are essential for effective and equitable health system governance.

Crisis response team

Across emergency response and crisis management contexts, these studies collectively emphasize that crisis effectiveness depends on coordinated, evidence-based, and institutionalized systems rather than isolated tools or reactive actions. Reviews by Alsaidan et al. (2024) and Akora et al. (2023) highlight the central role of inter-organizational cooperation, communication, and information-sharing—particularly through integrated governance structures and social media analytics—in improving preparedness, response, and recovery. Empirical studies on COVID-19 responses (Nikolić & Jagnjač, 2021) and emergency departments (Roth et al., 2025; Bailey et al., 2021) further demonstrate that structured teams, peer support, and data-driven dispatching enhance crisis outcomes when embedded in formal protocols. Technological and operational contributions, such as Khan et al.'s (2022) multi-UAV coordination model and Cruz et al.'s (2025) PREPaRE framework, reinforce that crisis tools are only effective when supported by training, leadership, and multidisciplinary collaboration. Finally, Sutarto et al. (2024) extend these insights to diaspora emergency planning, showing that inclusive, network-based

strategies strengthen response capacity in complex geopolitical contexts. Together, these studies frame crisis management as a systemic, collaborative, and continuously learned planning process, where coordination, leadership, and institutional integration determine policy and operational success. Schembourg et al. (2023) show that open dialogue in mental health crises evolves through interaction, enabling shared understanding and adaptive responses rather than fixed procedural control. Kim et al. (2024) demonstrate that reputational crises in sport can be mitigated through strategic leadership networks, where weak ties and symbolic authority help restore public trust during periods of instability. Similarly, White et al. (2024) highlight that frontline leaders in child protection units relied on reflective supervision, relational leadership, and emotional support to sustain decision-making capacity during the COVID-19 crisis. Together, these studies reinforce a broader insight across the literature: crisis effectiveness depends not only on formal plans and structures, but also on communicative practices, leadership relationships, and the capacity to adapt meaningfully under pressure.

Communication and Networking

Collectively, these studies demonstrate that crisis communication in hospitality and tourism is not an isolated operational activity but a strategically embedded, institutionally shaped, and digitally mediated planning function. Joseph et al. (2022) show that hotels' crisis communication during COVID-19 was strongly influenced by coercive regulations, professional norms, and mimetic pressures, leading organizations to converge around shared communication practices under uncertainty. Complementing this institutional perspective, Muruki et al. (2024) emphasize that communication effectiveness depends on pre-crisis preparedness, arguing that embedded communication strategies outperform reactive messaging. Several studies highlight the growing centrality of digital and social media platforms in crisis contexts. Silva and Silva (2023) and Carvalho et al (2023) demonstrate that social media and digital

communication function as core crisis management tools, enabling real-time engagement, reputation management, and organizational resilience when strategically coordinated. Empirical evidence from Spain further shows that hotels adapted message framing, tone, and platform use during COVID-19 to restore trust and stakeholder engagement (Fernández et al 2023). Using computational methods, Yu et al (2024) reveal that tourism crisis narratives evolve dynamically over time, underscoring the need for adaptive, data-driven communication strategies. Extending crisis communication impacts beyond immediate response, Liu et al. (2025) find that social media communication significantly shapes perceived service quality and tourists' behavioral intentions. Finally, Tai, Wang, and Xu's (2021) review states communication as a cross-cutting element across all crisis phases, while reinforcing Joseph et al.'s (2022) and Muruki et al.'s (2024) conclusions that communication is most effective when aligned with formal planning structures.Foundational perspectives on wireless communications are advanced by Zhang et al. (2025), who introduce the Mamba model as a scalable approach for handling long-range dependencies and spatiotemporal complexity in wireless networks, emphasizing efficiency trade-offs and future-ready architectures. Complementing this, Luo et al. (2025) demonstrate how AI reasoning—through large language models, autonomous agents, and cross-layer optimization—redefines wireless network intelligence, shifting communication systems from reactive data transmission toward adaptive decision-making frameworks. These AI-driven transformations are further extended by Guo et al. (2025), who propose generative semantic communication for mega-satellite networks, highlighting a paradigm shift from raw data transmission to meaning-based exchange to overcome bandwidth and latency constraints.

At the system optimization level, Wang et al. (2024) show that reinforcement learning has become central to communication optimization in complex underwater and 6G-integrated networks, particularly where uncertainty, energy efficiency, and quality-of-service constraints challenge traditional methods.

Similarly, Yu et al. (2025) address instability in microgrid communication by proposing hybrid long-distance networking technologies, demonstrating how intelligent optimization enhances reliability and security in geographically fragmented systems. Studies on autonomous and distributed environments reinforce these insights: Ma et al. (2025) emphasize that effective communication in unmanned underwater vehicle swarms depends on coordinated networking architectures capable of scalable control and data fusion, while Papuleti (2025) extends resilience thinking to interplanetary communication, proposing delay-tolerant architectures that ensure robustness under extreme latency and disruption.

At a broader infrastructural level, Ugwuumba (2025) situates these advances within the historical and architectural foundations of computer networking, highlighting how cloud computing, IoT, and security frameworks collectively redefine modern communication ecosystems. Finally, Saleem (2025) synthesizes these developments by demonstrating how AI networking enhances traffic management, fault detection, security, and automation across cloud, wireless, and 5G systems. Together, these studies converge on a shared conclusion: future communication systems are increasingly defined by intelligence, semantic efficiency, and adaptive optimization, where network performance and resilience depend not only on hardware capacity but on the integration of AI-driven reasoning, learning, and meaning-aware communication architecture. Zaidi and Mohd Yusof (2023) provide a foundational perspective by identifying core challenges in modern data communication and networking, particularly scalability, performance degradation, and reliability across heterogeneous devices and networks. This diagnostic framing complements more advanced studies by situating emerging technologies as responses to clearly defined structural limitations in current systems. Building on these challenges, Dallaf (2024) demonstrates how blockchain-based networking protocols offer a governance and security-oriented solution to long-standing issues of trust, privacy, and decentralization in communication networks. This work aligns with

broader trends in intelligent networking by reframing security not as an add-on feature but as an architectural property embedded within communication protocols themselves. When integrated with earlier AI- and optimization-focused studies, these contributions collectively illustrate a layered evolution of communication systems.

Crisis training

Crises highlight the need for systematic training to enhance preparedness, adaptability, and effective response across sectors, emphasizing learning, simulation, and continuous skill development as foundations of crisis management. Belias and Trihas (2022) establish that crisis situations in the hospitality sector necessitate systematic training for front-office employees, as operational changes during disruptions such as COVID-19 directly affect service delivery and employee confidence. Building on the importance of training identified at the employee level, Mahmoud et al. (2021) extend this perspective to hotel management. They demonstrate that training and development significantly improve managers' awareness of crisis management, showing a strong relationship between structured training programs and managers' ability to plan for, recognize, and respond effectively to crises. Similarly, Smart et al. (2021) highlight how learning and training supported hotels' responses to the COVID-19 pandemic. Their study shows that crisis coping strategies such as implementing health protocols, managing resources, and adjusting business operations—were closely linked to managerial learning processes. While the previous studies focus on hospitality, González-Arias and Laborda (2025) shift attention to the field of psychology, identifying a systemic crisis in psychologist training in Chile. They argue that deregulation and the absence of standardized training frameworks undermine professionals' capacity to respond to increasing social and psychological crises. Additionally, Shostatska and Kobylianskyi (2025) address crisis training within healthcare, emphasizing methodological and competency-based preparation. Expanding on training

methods, Evain et al. (2025) examine immersive crisis training using virtual reality. They argue that traditional crisis exercises are limited in realism and effectiveness, while immersive simulations enhance experiential learning, decision-making, and preparedness. Likewise, Minguez (2025) emphasizes continuity in crisis preparedness training at the institutional level. Her study presents a multi-annual training package that integrates regular simulations, internal and external coordination, and stakeholder collaboration. Furthermore, Mårtensson (2025) explores the use of generative AI-based virtual patient systems in crisis support training. In line with this technological focus, Knez (2025) provides an overview of crisis-support training in virtual reality, identifying research trends and gaps. Saryatmo and Sukhotu (2021), although focusing on digital supply chains, demonstrate that organizational performance during disruption depends on digital capabilities and knowledge development.

In the tourism and hospitality context, Bompolakis et al. (2022) demonstrate that tracing and addressing human capital educational needs enables hotels to strengthen crisis management capacity, particularly during COVID-19, by aligning training with evolving operational and employee demands. Similarly, Dey and Kumar (2025) emphasize that crisis management training in hotels improves preparedness, response, and recovery by equipping employees with procedural knowledge, communication skills, and adaptability under pressure. Beyond hospitality, crisis training in education and public institutions has gained attention through technology-enhanced approaches. Tyler (2025) shows that virtual reality–based crisis response training in education enhances experiential learning, collaboration, and decision-making skills, enabling educators and administrators to manage emergencies more effectively. This aligns with George (2024), who highlights that crisis management training in further education colleges is essential for institutional readiness, while also stressing that sustainable funding is necessary to maintain training effectiveness over time. In healthcare settings, crisis training is consistently

linked to improved clinical performance and decision-making. Yang and Kim (2025) demonstrate that simulation-based crisis resource management training significantly improves competencies among medical residents, confirming its value for emergency preparedness. Likewise, Babamiri and Karimiankakolaki (2025) find that crisis management training positively affects clinical decision-making and stress management among emergency medical service providers. Addressing systemic training challenges, Wang (2025) identifies organizational and interdisciplinary gaps in medical specialist training programs, emphasizing the need for structured crisis-oriented educational frameworks. Complementing this, Derkembaeva (2025) shows that simulation trainers and models enhance cognitive strategies, situational awareness, and performance of anesthesiology and resuscitation professionals during crisis situations. Finally, Dorgan et al. (2024) reveal that inadequate specialty training and workforce instability undermine crisis response capacity among resident doctors, reinforcing the importance of structured and supportive crisis training pathways.

Crisis response practices

Crisis response practices are increasingly understood as interconnected organizational and institutional processes rather than isolated emergency actions. Pandya and Boukareva (2022) argue that effective crisis response begins with innovative human resource practices that safeguard employee well-being while ensuring business continuity, positioning HR as a frontline crisis function. This human-centered foundation is reinforced by Mukolwe. (2023), who found that training, supportive leadership, and employee involvement significantly improve hotel performance during crises. Translating internal preparedness into operational action, Le et al. (2021) emphasize adaptive operations, digital transformation, and proactive communication as essential crisis response practices for sustaining service delivery and rebuilding customer trust. From a public systems perspective, Afrihyia et al. (2025) highlight

that crisis response effectiveness depends on early warning systems, coordinated governance structures, and well-trained response teams, noting that fragmented coordination weakens response capacity. In humanitarian health settings, Halder and Hasan (2023) demonstrate that crisis response must integrate gender and protection mainstreaming to ensure inclusive, ethical, and effective services for vulnerable populations. Complementing this, Svenbro (2022) shows that calculative practices such as data, statistics, and metrics are central to organizing and monitoring crisis responses, though excessive reliance on numbers may overlook social consequences. Expanding further on refugee crisis management, Halder et al. (2024) emphasize that crisis response practices must balance operational efficiency with protection mechanisms to prevent marginalization and ensure human dignity in emergency contexts. Finally, Magsi et al. (2023) find that artificial intelligence strengthens crisis response accuracy and speed only when guided by emotionally intelligent leadership and structured change management practices.

Operations

Operations in crisis contexts are consistently framed as systems requiring preparedness, adaptability, and coordination across sectors. Sharma et al. (2021) emphasize that tourism and hospitality operations must integrate disaster risk reduction into core management structures, arguing that resilience and recovery depend on proactive planning, stakeholder coordination, and continuity mechanisms embedded in daily operations. From a strategic operations lens, Zakharova and Strekalov (2025) highlight that crisis management effectiveness depends on personalization, cost optimization, and innovation. Focusing on public-sector operations, Rahman (2023) demonstrates that governance failures during large-scale catastrophes stem from weak operational coordination and delayed decision-making. In organizational settings, Hidayat and Wahyuni (2023) argue that crisis operations are shaped by organizational culture and leadership. Their study of PT Nindya Karya shows that operational effectiveness

during crises relies on institutional values. Supply chain operations receive extensive attention in crisis contexts. Sanni (2025a) explains that logistics operations during disruptions require entrepreneurial responses, emphasizing flexibility, customer-focused service continuity, and rapid operational reconfiguration. Complementing this, Sanni (2025) stresses that risk communication is integral to logistics operations. Note on future-oriented operations is provided by Zeimpekis and Madas (2025), who argue that resilient supply chain operations must integrate sustainability, digitalization, and innovation. In the hospitality and legal domain, Green and Varburgh (2021) emphasize that operational decision-making during COVID-19 depended heavily on legal literacy. Healthcare operations during crises are examined by Paul (2024), who identifies structured communication, delegation, and accountability as essential operational principles. Xia (2025) demonstrates that corporate operations during brand crises rely on coordinated communication strategies, rapid response, and cross-functional alignment. Liu et al. (2024) argue that innovation-driven operational initiatives, digital tools, coordination mechanisms, and adaptive service delivery strengthen community resilience during crises. Brynzylo (2025) demonstrates that operational security in shopping and entertainment centres depends on integrated crisis plans, shelters, communication systems, and staff training under armed aggression. Lastly, Marjori (2026) shows that hotel operations rely on preparedness, employee actions, and emergency protocols, though gaps remain in communication roles during crises.

1.4 Schematic Diagram

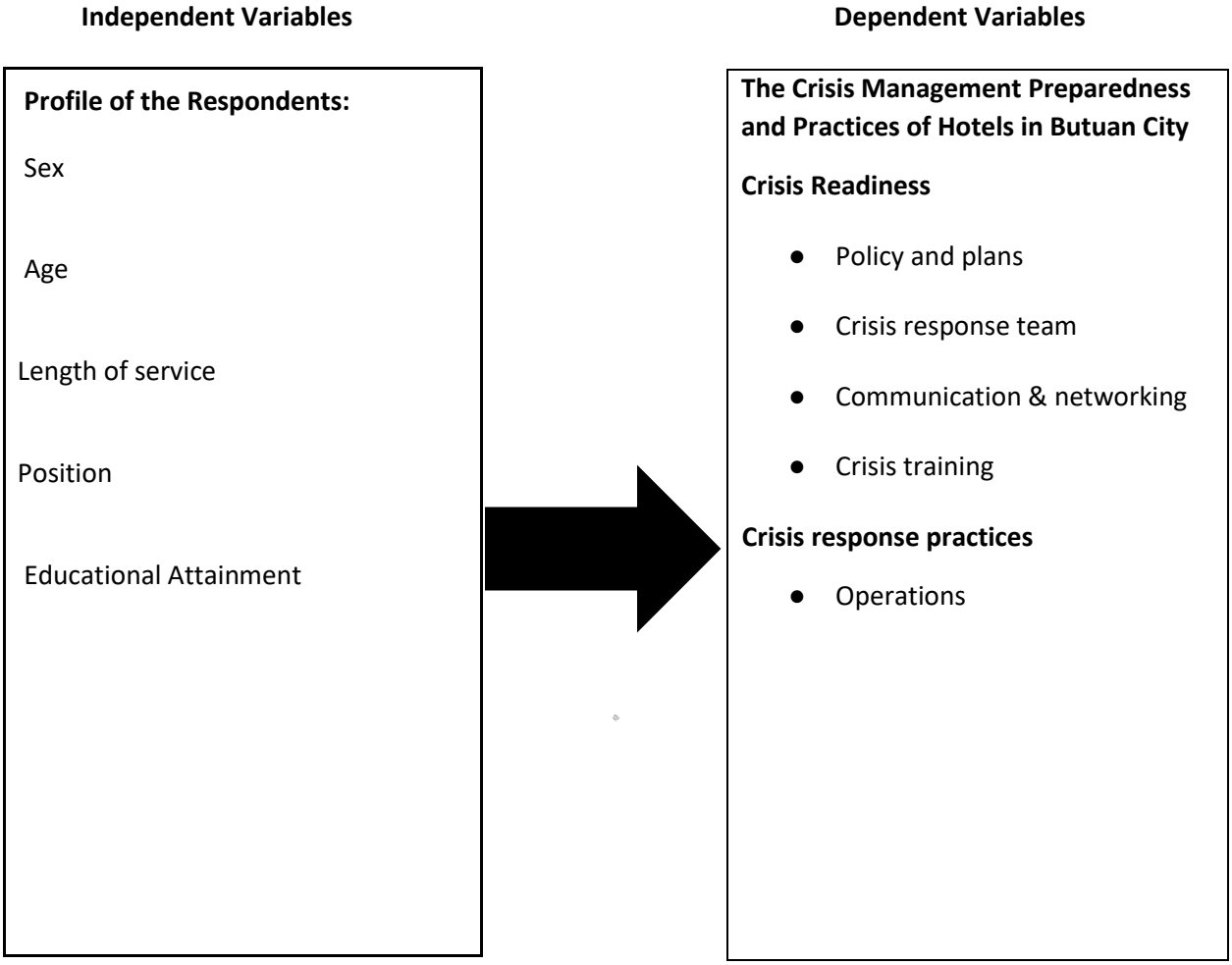


Figure 1. The Schematic Diagram shows the Interaction of the independent variables and dependent variables of the study.

1.5 Statement of the Problem/ Objectives of the Study

This study will aim to determine the crisis management preparedness and practices of hotel staff in Butuan City, Agusan Del Norte. Specifically, it will seek to answer the following questions:

1. What is the profile of the respondents in terms of?

    1.1 Sex;

    1.2 Age;

    1.3 Length of Service;

    1.4 Position; and

    1.5 Educational Level?

2. What is the level of perception of hotel staff receptionists in preparedness in terms of?

    2.1 Policy and plans;

    2.2 Crisis response team;

    2.3 Communication and networking; and

    2.4 Crisis training?

3. What is the level of Crisis management practices of hotel staff in selected hotels in Butuan City in terms of?

    3.1 Operations?

4. Is there a significant relationship between the hotel staff's perception of crisis management and their practices?

1.6 Hypothesis

There is a significant relationship between the hotel staff's perception of crisis management and their practices.

1.7 Scope and Limitation

The study shall be concerned with the hotel employees and guests in Butuan City, Agusan del Norte, exploring the demographic and professional profiles of employees along with their age, gender, length of service, position, and educational background, as well as the preparedness and effectiveness in crisis management. In the case of the guests, it shall tap experiences, perceptions, and satisfaction relating to how hotels deal with crises. The study is delimited to selected hotels within Butuan City. The data shall depend on the self-reporting of the respondents, which may involve personal bias in responding, and does not include tourism establishments outside the area under study.

1.8 Definition of Terms

To provide clarity and a better understanding of the study, the following terms are defined conceptually and operationally.

Age. It refers to the ages of employees working in the hotel each year. Age is considered a statistical factor that may influence the level of preparedness and crisis management practices among hotel employees.

Crisis management plan. Is a formal document that outlines procedures and responsibilities, as well as communication protocols, in response to emergencies.

Crisis prevention. Within this context involves strategies and policies to reduce or eliminate the risk of an operational disruption, guest harm, or loss to business in hotel establishments.

Crisis response. Can be explained as all the actions hotel management and staff do in order to minimize damages, ensure safety, and keep operational functionality.

Crisis Response Team. Is a preidentified and specialized group of hotel personnel who make decisions, execute emergency procedures, and coordinate actions during any crisis event.

Educational attainment. It refers to the level of staff education in the hotel industry.

Emergency communication. Is the timely and effective dissemination of information during a crisis to guests, staff, authorities, and other relevant stakeholders in written, verbal, or digitized format.

Emergency procedures. Define a standard action and safety measures whenever crises arise, such as guest evacuation, fire response, medical assistance, and lockdown procedures.

Guest. A guest is any person who utilizes accommodation, lodging, or any other service made available by a hotel for their leisure, business, or other purposes.

Hotel staff/ Hotel employees. It refers to individuals who work in hotels and are responsible for managing the day-to-day operations and resolving issues. This study primarily focuses on front desk receptionists and their readiness for various situations, as well as how they handle them.

Operational disruption. This implies the loss of power, reduction of guest services, damage to or loss of facilities, and personnel shortages brought about by crises that hamper hotel operations.

Pre-crisis preparedness. Is the proactive measures that hotels initiate to anticipate and prevent emergencies by planning, training, and resource allocation before the actual crisis.

Risk Assessment. It is a procedure used to identify, analyze, and evaluate the potential hazards or vulnerabilities that would affect hotel operations, staff, and guests during a crisis.

Sex. It refers to the biological classification of the respondents as male or female. The study aims to determine whether gender influences the different perspectives and practices of crisis management among hotel staff.

Staff emergency training. Is the organized programs offered to employees at the hotel with a view to equipping them with the knowledge and skills necessary for dealing with crises, including evacuation, first aid, and security responses.

Threat monitoring. Involves observing and evaluating events both internally and externally that might pose a threat to the safety of the hotel, such as weather alerts, guest incidents, and security reports.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter will discuss the research design, materials and methods, population and sample, and the instruments and procedures employed. It will also describe the data collection and analysis procedures, as well as the ethical considerations used in the study.

2.1 Research Design

The research design for this study is descriptive-correlational. This research will explore the readiness of hotels in Butuan City regarding natural calamities, civil disturbances, pandemics, and technical issues and their crisis management strategies. It shall contain full subsections describing the preparedness of both the hotel employees and the experiences and perceptions of guests during emergencies. For the preparedness of hotel staff, it will look into preparedness, communication, cooperation, and recovery actions. In the case of guests, it will look into the awareness of safety measures, confidence in the hotel's response, and satisfaction with the handling of a crisis. These variables will be analyzed to determine whether there are significant relationships between the preparedness of the staff and guest perceptions regarding the effectiveness of crisis management.

2.2 Research Materials and Methods

This study will be using primarily a structured questionnaire to collect quantitative data from both hotel employees and guests about crisis preparedness and practices. The research instrument is anchored on the study of Cabaguing et al. (2024). However, certain variables included in the original instrument were excluded, since these did not conform to the specific objectives and scope of the present study. The items of the questionnaire will be adopted and modified from past studies and existing tools within the hotel industry that address crisis-related concerns. The survey tool will be modified to address both

groups properly, and it is divided into three main sections. Section 1 (Demographic Profile) describes participants' sex, age, and highest educational attainment; for hotel employees, years of service and job position; and for guests, their frequency of hotel stays. Section 2 will assess the crisis readiness among hotel employees and the safety awareness of guests in crisis situations. Section 3 will assess disaster response practices in terms of proper communication, cooperation, and recovery strategies, based on operational roles of employees and experiences or expectations by the guests.

Response categories range from 1 = Strongly Disagree to 4 = Strongly Agree. The revised questionnaire will be content-validated by experts in hospitality management and research methodology for clarity, accuracy, and relevance of the items. A pilot test will also be conducted to establish the reliability or internal consistency of the instrument. Data presentation, analysis, and interpretation will use appropriate statistical tools.

Frequency and Percentage: These will be used to answer SOP 1, which aims to present the demographic profile of both hotel employees and guests.

Mean: This will be used to answer SOPs 2 and 3, which focus on assessing perceptions of how well-prepared hotel employees are in crisis response and how well guests perceive the ability of the hotel to respond appropriately to emergencies.

Correlation: This will be used to answer SOP 4, which seeks to ascertain whether there is a significant relation between employees’ perceived crisis preparedness and response, and guests’ perceptions of how effectively the hotel manages crisis situations.

2.3 Research Environment, Population and Sample

The study will be conducted in Butuan City, Agusan Del Norte, located in the Caraga Region (Region XIII) in the northeastern portion of Mindanao, Philippines. Butuan City is approximately 24.3 kilometers from Saint Michael College of Caraga and is geographically located in the Northern East part of Mindanao. Butuan City has a land area of 816.62 square kilometers (315.30 square miles). The population was 385,530 according to the 2024 Census. Butuan will be recounted as a salient pre-colonial city-state in contemporary northeast Mindanao, ruled by a Rajah known for its Indianized political organization, metallurgical economy, and advanced naval engineering, as well as for having active diplomatic ties with China in 1001 CE. Archaeological finds include balangay boats from Libertad, a community renowned for its gold, and artifacts such as deformed skulls that reveal the richness, skill, and vibrancy of the village, where people, goods, and ideas populated. Butuan City has a postal code of 8600 and is made up of 86 barangays. Being a highly urbanized regional center, Butuan City provides a realistic setting for gauging the preparedness and crisis response measures of hotels. This will be the location for assessing the readiness and practices of crisis management; respondents will include both hotel employees and guests. Employees are able to provide insight into operational procedures and preparedness, while guests can provide perceptions and experiences regarding the hotel's crisis response.

The respondents of this study will be selected hotel employees and guests of hotel establishments in Butuan City. Hotel employees directly involved in the operation of the hotel, including the response plan for crises, including the likes of the front office, housekeeping, food and beverage service, and security personnel, will participate in the study. However, only hotel employees who have served at least six (6) months of continuous service at the hotel will participate in the study, thus giving them adequate experience with the standard operations of the hotel.

The sampling of the hotel's employees will be conducted using purposive sampling since the study is focusing on the employees who have the necessary knowledge and are involved in the related operations during the occurrence of the crisis. On the other hand, the sampling for the hotel guests will be conducted using convenience sampling due to their availability. The involvement of the hotel guests in the study will aid in the study by providing an insight from the guests on their experience and observations about the way the hotels have managed the crisis.

2.4 Research Data and Analysis Plan

The data will be processed in phases to ensure that the results of the study are accurate, reliable, and meaningful. All quantitative data will be coded, tabulated, and organized for statistical analysis. Descriptive statistics like frequency, percentage, mean, and standard deviation will describe the demographic profiles in both hotels for employees and guests and present an analysis of the perceptions of crisis preparedness and response. The Pearson Product-Moment Correlation Coefficient (r) is used to determine if there is a significant relationship between the perceived crisis preparedness of employees and their actual crisis response, and between the perception of crisis management by the guests and the preparedness level of the hotel. A 0.05 level of significance shall be considered in all statistical tests. The research study, therefore, solely considers quantitative approach and methods to quantify the observations, ensuring objectivity in measurement and numerical interpretation to allow for the results to reflect the preparedness of the employees and responsiveness to states of crisis and the perceptions of the guests on the aspects of crisis management in Butuan City hotels.

2.5 Research Ethical Standards

This research, "The Crisis Management Preparedness and Practices of Hotels in Butuan City", follows the ethical principles that guide the protection of participants' rights, data privacy, and research integrity.

A. Protection of Intellectual Property Rights (IPR)

Every source of information, ideas, and data that has been used in this research will be credited accordingly via citations and references to avoid plagiarism. The researcher will ensure that all resources, both print and digital, comply with the intellectual property rights of the original authors.

B. Informed Consent

Participants will be thoroughly informed about the research goals, purpose, and procedures before participating. A consent form will be provided to ensure that respondents voluntarily participate and understand their rights as respondents, and to prevent any respondent from being forced or pressured to take part in the research.

C. Data Privacy and Confidentiality

The confidentiality pertaining to all the information gathered from the research study will not be broken by the researchers. There will be no exposure or sharing of personal data about the individuals and the hotels involved without their consent. The data will be kept in a secure place and accessed only for scholarly purposes, and participants' privacy and professional reputations will be reassured. The names of the people who took part in the survey will not be mentioned in any part of the research paper.

D. Voluntary Participation and Freedom to Withdraw

Participation in the study will be entirely voluntary. Respondents may withdraw or refuse to participate without facing any adverse consequences.

E. Minimization of Harm and Risk Management

The researcher will ensure that research participants are not harmed or discomforted, whether physically, emotionally, or in their reputation. The study will be conducted with full consideration of their well-being while ensuring a safe and respectful research environment.

F. Beneficence and Contribution to Knowledge

The purpose of this research is to provide hotels in Butuan City with novel ideas on how to successfully manage and respond to emergencies, thereby contributing to the current body of professional knowledge in the hospitality industry.

G. Justice and fair Participant Selection

Participants will be selected in accordance with their relevance to the research, and the process will be conducted fairly. There will be no discrimination according to gender, role, or association, and the researchers will exercise care to avoid any form of prejudice or unfair treatment when selecting participants for representation.

H. Data integrity and Accuracy

The information that the researchers were able to gather was not fabricated, changed, or manipulated in any way. The data were completely accurate, and they had been recorded, and the researchers reviewed all data carefully to ensure it was analyzed and appropriately gathered, according

to all protocols that were applicable. To ensure the study remains valid, the results must be reported accurately and reliably.

I. Transparency and Honesty in Reporting

The study's findings are presented clearly and directly, with equal emphasis on both the positive and negative elements of the research. To obtain a different result, there will be no lying, altering, or concealing of the facts. Everyone will be able to view the entire study, from the beginning stages of data collection to the end stages of data analysis and the dissemination of the findings.

J. Use of Patented or Copyrighted Materials

Researchers must not use copyrighted or proprietary content without permission or proper citation. You need permission from the proprietors to utilize specific types of content, like frameworks and questionnaires. The researchers will get all the permissions they need and follow all the copyright restrictions that apply.

K. Ethical Considerations for Animal and Human Trials

The researcher will make certain that all the participants are treated with respect and that their rights and dignity are respected throughout the entire process of conducting this research project, which only involves human volunteers. In every instance, involvement will be conducted in accordance with ethical and moral standards.

L. Responsible Use of Ai and other Related Technologies

The use of artificial intelligence tools, such as Grammarly, Quillbot, Elicit, and ChatGPT, is of utmost significance to our investigation because they facilitate the research process. Now that we have

improved and clarified our writing, we can identify and correct any errors that may have been made. You can generate ideas, summarize them, and organize them into papers and surveys with the assistance of ChatGPT. We will gather information and write our documents with the help of Elicit, which also assists us in summarizing important points and presenting the results of our study in a clear and easy-to-understand manner. The inclusion of each of these will result in an improvement to our query. Our investigation will be improved by the incorporation of each of these technologies, which will function as supplementary components. The specialists will accept complete responsibility for all consequences arising from their work, to preserve the authenticity and originality of the circumstances.

M. Ethical Clearance and Institutional Approval

Taking these steps will help ensure that the researcher complies with all relevant legislation regarding the protection of respondents, data privacy, and intellectual property rights. The researcher will get ethical permission from the Institutional Review Board of Saint Michael College of Caraga.